

Siwei Chen

📍 Beijing, China ✉ 2300017751@stu.pku.edu.cn ☎ 18810613228 🌐 pkuchensw

Personal Statement

I am currently a junior at Yuanpei College, Peking University, majoring in Data Science and Big Data Technology. I am deeply interested in Artificial Intelligence and motivated by the rapid development of large language models. My research interests focus on efficient LLM algorithms and systems, especially post-training methods such as reinforcement learning, as well as emerging efficient model architectures. With a solid foundation in mathematics and programming, I enjoy working on problems that require both theoretical understanding and practical system design. My current research explores how algorithm-system co-design can make large language models more efficient, reliable, and deployable.

Education

Yuanpei College, Peking University

Sept 2023 - Present

B.S. in Data Science and Big Data Technology;

- GPA: **3.8/4.0**; Major Rank: 1/13; Overall Rank: 43/355
- **Basic Courses:** Machine Learning (99), Practice of Programming in C&C++ (98), Data Structure and Algorithm (A) (95), Probability theory and statistics in information science (95), Advanced Algebra (93)
- **Advanced Courses:** Programming in Artificial Intelligence (99), Principles, Challenges, and Industrial Applications of Large Language Models (99), Directed Research in AI System (95)
- **Teaching Assistant:** Probability theory and statistics in information science, Outdoor Exploration

Tsinghua University High School

Sept 2017 - Jun 2023

High School Certificate;

- College entrance examination: 693/750

Research Experience

Efficient Reinforcement Learning for Large Language Models

May 2025 - Present

Advisor: Prof. Bin Cui and Prof. Xupeng Miao, Peking University

- Investigated efficiency bottlenecks in LLM reinforcement learning, including the straggler effect caused by long-tail rollout lengths and the length explosion problem in LLM RL training.
- Proposed DARTS (Distribution-Aware Active Rollout Trajectory Shaping), a system-level optimization paradigm that actively reshapes the model's output distribution to reduce intra-prompt redundancy. Designed Dual-End Sampling, Adaptive Redundant Allocation, and a token-level streaming mechanism to accelerate RL training while preserving reasoning capability.
- Proposed QLPO, a resampling-based variant of GRPO that introduces implicit length control by favoring short-correct and long-incorrect trajectories without modifying the reward function.

Efficient Diffusion Language Models

Feb 2026 - Present

Advisor: Prof. Fan Lai, University of Illinois Urbana-Champaign

- Exploring diffusion language models for efficient parallel generation, focusing on improving the quality-efficiency trade-off beyond standard autoregressive and block-diffusion decoding.
- Investigating scalable training and inference designs that enhance self-consistency, long-block generation, and practical serving efficiency.

Context Management Benchmark for Long-Horizon LLM Agents

Feb 2026 - May 2026

Advisor: Prof. Bin Cui and Prof. Xupeng Miao, Peking University

- Worked on MemClaw-Bench, a capability-centered benchmark for evaluating context management in long-horizon tool-using agents under compressed-context settings.
- Designed diagnostic tasks and evaluation protocols covering memory capabilities such as constraint tracking, version updating, procedure transfer, source conflict resolution, task resumption, and staleness judgment.

Publications

DARTS: Distribution-Aware Active Rollout Trajectory Shaping for Accelerating LLM Reinforcement Learning (ICML 2026) May 2025 - Jan 2026

Yujie Wang†, *Siwei Chen*†, Longzan Luo†, Xinyi Liu, Xupeng Miao, Fangcheng Fu, Bin Cui (Co-first author)

arxiv.org/abs/2605.30859 

QLPO: Quadrant-weighted sampling for Length-aware Policy Optimization Dec 2025 - March 2026

Siwei Chen, Siqi Chen, Xupeng Miao, Bin Cui (First Author)

MemClaw-Bench: A Capability-Centered Benchmark for Context Management in Long-Horizon Agents (Under Review) Feb 2026 - May 2026

Siwei Chen, Xupeng Miao, Bin Cui (First Author)

Beyond Sight: Towards Cognitive Alignment in LVLM via Enriched Visual Knowledge (CVPR 2025) Sept 2024 - Dec 2024

Yaqi Zhao, Yuanyang Yin, Lin Li, Mingan Lin, Victor Shea-Jay Huang, *Siwei Chen*, Weipeng Chen, Baoqun Yin, Zenan Zhou, Wentao Zhang

arxiv.org/abs/2411.16824 

Projects

Usagi: ICS course helper

[Usagi project](#) 

- Developed a Retrieval-Augmented Generation enhanced LLM question-answering framework. Backed by a comprehensive corpus of learning materials, the system supports RAG-based source tracing, predictive question generation, and learner cognitive state diagnosis. Over 100 students have used our system and given positive feedback.
- Tools used: Python, Vibe coding for front-end

MyTorch: Custom PyTorch for CNN training

[MyTorch](#) 

- Implemented a custom PyTorch based on CUDA programming. My implementation includes basic kernels, encapsulation between Python and C, and the construction of a simple computational graph. The project can ultimately train a CNN for image recognition.
- Tools used: CUDA, C++, Python

Honors and Awards

- 2023 Beijing Merit Student
- 2023-2024 Peking University Freshman Scholarship
- 2023-2024 Peking University Second-Class Scholarship
- 2023-2024 Peking University Merit Student
- 2024-2025 Peking University third-Class Scholarship
- 2024-2025 Peking University Outstanding Research Award

Additional Information

Programming Languages: Python, C++, Rust

Tools: PyTorch, CUDA

Language: New TOEFL iBT: 5.5/6 (Reading 6.0, Listening 5.0, Writing 5.0, Speaking 5.0)

I enjoy all kinds of sports ranging from basketball, running to Ultimate Frisbee and mountain climbing. I am a member of the Shanying Club (Mountaineering Association of Peking University) and I won the second prize at the PKU Ultimate Frisbee Cup (2024,2025).